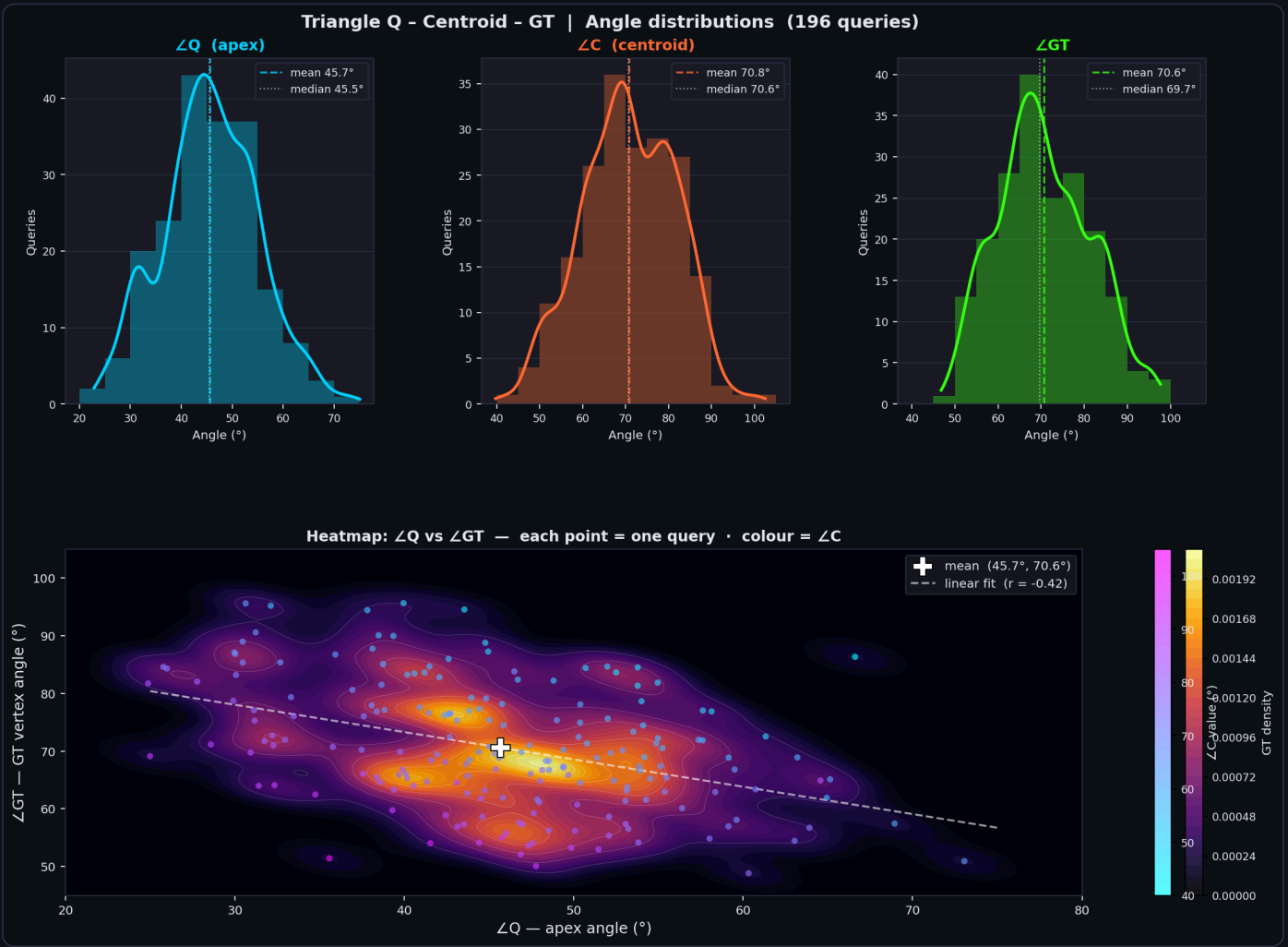
Triangle Geometry: Q — Centroid — GT

196 queries · angular distances (spherical) · G3 centroid = oracle (LLM score = 1.0)

Angle distributions & heatmap (∠Q vs ∠GT)



Top row: histogram + KDE for each vertex angle (∠Q, ∠C, ∠GT) separately.
Bottom: 2D heatmap — each point is one query plotted by its (∠Q, ∠GT) pair. Colour encodes ∠C. The negative correlation (r = −0.42) is visible as a downward trend: when ∠Q is large, ∠GT tends to be smaller, consistent with the near-constant angle sum (≈187°).

Angle distributions — count per 10° bin

∠Q — apex angle

Range	n	%	Bar
[20–30)°	8	4.1%	
[30–40)°	44	22.4%	
[40–50)°	**80**	**40.8%**	
[50–60)°	52	26.5%	
[60–70)°	11	5.6%	

∠GT

Range	n	%	Bar
[50–60)°	33	16.8%	
[60–70)°	**68**	**34.7%**	
[70–80)°	53	27.0%	
[80–90)°	34	17.3%	
[90–100)°	7	3.6%	

Range	n	%	Bar
[70–80)°	1	0.5%	

∠C given ∠GT bin

> Pearson r(∠GT, ∠C) = **−0.668**. When ∠GT is sharp (small), the centroid angle is larger — the centroid deflects further from the GT direction.

∠GT bin	n	∠C mean	∠C std	∠Q mean
[50–60)°	33	**81.5°**	8.0°	50.5°
[60–70)°	68	74.4°	7.5°	47.6°
[70–80)°	53	67.1°	8.5°	44.5°
[80–90)°	34	61.5°	10.0°	40.6°
[90–100)°	7	56.6°	6.3°	36.3°
All 196	196	70.8°	9.3°	45.7°

Full distribution table (sides + angles)

Quantity	Unit	Mean	Median	Min	Max	p10	p90
Sides — Cosine distance							
Q → GT	—	0.17	0.15	0.06	0.65	0.10	0.26
Q → C	—	0.17	0.15	0.07	0.48	0.11	0.25
C → GT	—	0.10	0.09	0.02	0.47	0.05	0.16
Sides — Euclidean distance							
Q → GT	—	0.58	0.56	0.35	1.14	0.44	0.72
Q → C	—	0.57	0.55	0.38	0.98	0.46	0.70
C → GT	—	0.43	0.41	0.22	0.97	0.31	0.57
Angles — Cosine / Spherical							
∠Q (apex)	°	**45.7**	45.5	24.9	73.1	32.2	57.6
∠C (centroid)	°	70.8	70.6	41.8	100.5	56.5	84.8
∠GT	°	70.6	69.7	48.8	95.7	56.6	84.9
Angles — Euclidean / Flat							
∠Q (apex)	°	**43.6**	43.8	21.7	69.4	30.6	54.2
∠C (centroid)	°	68.2	68.0	37.6	96.3	55.2	81.6
∠GT	°	68.1	67.7	46.8	93.5	54.4	82.3